

Formal WiFi Substrate Observation Guide

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ABSTRACT

This paper describes a method of using WiFi Received Signal Strength Indication, commonly referred to as RSSI, to observe and document disturbances in environmental electromagnetic fields. The method provides a structured framework for detecting and recording changes in the WiFi signal propagation environment that correspond to the introduction, movement, or removal of physical objects or biological mass within the signal path. The approach is non-invasive, requiring no modification to the subject or surrounding environment beyond the placement of standard consumer-grade networking hardware. The procedure is fully repeatable under controlled conditions and relies exclusively on consumer-grade devices available to the general public, making it accessible for use without specialized laboratory equipment or proprietary instrumentation.

INTRODUCTION

The purpose of this document is to establish a formal, reproducible methodology for using WiFi RSSI values as a detection mechanism for environmental field disturbances. WiFi radio frequency signals, operating in the 2.4 GHz and 5 GHz bands, propagate through the surrounding environment in ways that are directly influenced by the physical composition of that environment. Any object, material, or biological mass that intersects or borders the signal path will alter the signal in measurable ways, including attenuation, reflection, and scattering of the transmitted wave.

RSSI is a numerical metric reported by WiFi client devices that represents the relative strength of the received signal from a given access point or router. Because RSSI is sensitive to physical changes in the propagation environment, it serves as a practical and accessible

proxy for detecting disturbances that would otherwise require specialized electromagnetic sensing equipment to identify. Even modest changes in the physical environment near a WiFi transmitter or receiver produce detectable changes in RSSI values.

This method is intended to provide a structured, reproducible approach to documenting anomalous field behavior. By establishing a stable baseline and then introducing controlled variables, the observer can distinguish genuine environmental field disturbances from ordinary ambient noise. The result is a systematic record of signal behavior that can be analyzed, compared, and shared in a professional context.

EQUIPMENT

To conduct this observation, the following devices and tools are required. A WiFi-enabled router or wireless access point capable of broadcasting on either the 2.4 GHz or 5 GHz band is necessary to serve as the signal source. A WiFi-enabled client device, such as a laptop computer, smartphone, or tablet, is required to receive the signal and report RSSI values. RSSI monitoring software or a dedicated application must be installed on the client device to display real-time signal strength readings at a sufficient sampling rate for observation purposes.

A notepad, data logging application, or spreadsheet program is required to record RSSI values and observations over time. A measuring tape or ruler is needed to establish and document consistent distances between the router, the client device, and any introduced objects or subjects. A stable mounting surface or adjustable tripod is recommended for securing the router at a fixed position throughout the procedure to eliminate unintended movement as a variable.

BASELINE BEHAVIOR

Under stable conditions, with no objects moving near the router or the client device and no persons or animals entering or exiting the observation area, RSSI values remain relatively consistent within a narrow variance range. A typical stable RSSI reading for a consumer WiFi

setup at short to medium range will fluctuate within a window of approximately two to four decibel-milliwatts, depending on the hardware in use and the ambient radio frequency environment of the location.

Minor fluctuations are considered normal and are attributable to environmental noise, including interference from neighboring wireless networks, minor thermal variation in hardware components, and low-level multipath effects inherent to any enclosed space. These baseline fluctuations do not follow a directional trend and do not correlate with any observable physical event. A stable baseline reading is characterized by values that oscillate within a narrow, predictable range over an extended sampling period, with no sustained upward or downward drift and no abrupt departures from the established variance window. This baseline profile serves as the reference against which all subsequent observations are compared.

PROCEDURE

The first phase of the procedure involves establishing the baseline. Begin by positioning the router on a stable mounting surface or tripod at a measured distance from the client device. The recommended starting distance is between one and three meters, adjusted as needed for the specific environment. Power on both devices and allow the WiFi connection to stabilize for a minimum of five minutes before recording begins. Once the connection is stable, begin recording RSSI values at regular intervals, capturing at least one reading every ten seconds. Continue recording for a minimum of ten minutes with no persons or objects moving within the observation area. Calculate the mean and variance of the recorded values to define the baseline profile for this session.

The second phase involves the introduction of a test object or subject. While the RSSI monitoring software continues to record values, introduce a test object or subject into the WiFi signal path. The object should be positioned at a defined point between the router and the client device, or adjacent to the router, as specified by the observation design. Record the RSSI values immediately following introduction and continue recording for a sufficient

duration to characterize the signal change. Note the timing, magnitude, and direction of any observed change in RSSI relative to the established baseline.

The third phase involves movement verification. With the test object or subject remaining within the observation area, move it slowly and deliberately through different positions within or adjacent to the signal path. Observe and record the corresponding changes in RSSI values as the object changes position. Document the relationship between the direction and distance of movement and the observed RSSI change. This phase confirms that the signal changes are responsive to the physical location and motion of the introduced variable rather than being artifacts of the recording system.

The fourth phase is the control condition. Remove the test object or subject entirely from the observation area and allow the RSSI readings to return to the baseline range previously established. Wait until the values have stabilized within the original variance window before concluding this phase. Record the duration required for the signal to return to baseline and confirm that the final stable values fall within the expected range. This step verifies that the observed disturbances were caused by the introduced variable and that the system behaves reproducibly in its absence.

The fifth phase is repetition. Repeat the full procedure, including baseline establishment, object introduction, movement verification, and control condition, a minimum of three times within the same session. Average the recorded RSSI change values across all repetitions to produce a mean disturbance magnitude for the session. Document any variation between repetitions and note environmental conditions that may have differed. Consistent results across repetitions constitute strong evidence that the observed signal behavior is reliably linked to the physical variable under observation.

EXPLANATION

WiFi signals are a form of electromagnetic radiation that propagates through space as radio frequency waves. These waves interact with the physical environment through four primary

mechanisms: absorption, reflection, scattering, and diffraction. Each of these mechanisms produces a measurable change in the signal as received by the client device, and together they account for the sensitivity of RSSI to physical changes in the signal path.

Absorption occurs when a material converts electromagnetic wave energy into heat rather than allowing it to pass through or reflect it. Materials with high water content are particularly effective absorbers of both 2.4 GHz and 5 GHz signals because water molecules interact strongly with electromagnetic radiation at these frequencies. The human body, which is composed largely of water, is one of the most effective absorbers of WiFi signal energy encountered in ordinary environments. When a person or any object with significant water content enters the signal path, a measurable drop in received signal strength is expected.

Reflection occurs when a signal encounters a surface and is redirected rather than absorbed or transmitted. Dense materials such as metal, concrete, and certain ceramics are effective reflectors of WiFi signals. When a reflective object is introduced into the signal environment, it can either increase RSSI at the client device by redirecting signal energy toward the receiver, or decrease it by redirecting energy away from the receiver, depending on the geometry of the situation. This means that RSSI changes caused by reflective objects may manifest as increases rather than decreases in signal strength.

Scattering occurs when a signal encounters an irregular surface or an object whose dimensions are comparable to the wavelength of the signal, causing the signal energy to be dispersed in multiple directions. This reduces the coherent signal energy reaching the client device and typically manifests as a reduction in RSSI accompanied by increased variance in readings. Diffraction, the bending of a signal around obstacles, is also relevant at these frequencies and can cause interference patterns that produce rapid, location-sensitive fluctuations in signal strength.

The combined result of these propagation effects is that RSSI is genuinely sensitive to the physical state of the environment through which the WiFi signal passes. Objects or biological

mass introduced into or moved through the signal path reliably produce observable changes in RSSI values. The 2.4 GHz band, having a longer wavelength, penetrates soft materials more effectively but is more affected by reflective surfaces. The 5 GHz band, with its shorter wavelength, is more readily absorbed by organic material and attenuates more quickly over distance. Both bands are suitable for this observational method, though the specific magnitude of the observed effects will differ between them.

SUMMARY

This document presents a formal observational method for detecting and documenting environmental field disturbances using the Received Signal Strength Indication of a standard consumer WiFi network. The method is grounded in the well-established physical behavior of electromagnetic wave propagation and requires no specialized equipment beyond devices commonly available to the general public.

The core of the method involves establishing a stable signal baseline under undisturbed conditions, introducing a controlled physical variable into the signal environment, observing and recording the resulting changes in RSSI, and then verifying that the signal returns to baseline when the variable is removed. Repeating this process multiple times and averaging the results produces a robust, defensible record of the observed phenomenon.

The physical basis for the method is straightforward. WiFi signals are electromagnetic waves that are absorbed, reflected, and scattered by physical objects and biological mass. The human body and objects with high water content are particularly effective at attenuating signals in the 2.4 GHz and 5 GHz bands. These interactions produce measurable changes in the RSSI values reported by a standard client device, making consumer WiFi hardware a practical instrument for structured environmental observation.

The method described in this guide is non-invasive, repeatable, and accessible. It is suitable for use by any observer who can operate standard consumer networking equipment and run a basic RSSI monitoring application. The results it produces are quantitative, time-stamped,

and directly tied to observable physical events, making them appropriate for inclusion in a formal observational record or technical report.